

Smoking Cessation, Weight Change, and Risk of Dementia

A Prospective Cohort Study

Hui Chen,^{1,2,*} Jingjing Wang,¹ Sirui Lai,¹ Guoping Peng,¹ Geng Zong,³ Changzheng Yuan,^{2,4} and Benyan Luo^{1,5,*}

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Correspondence

Dr. Chen
hui.chen@zju.edu.cn
or Prof. Luo
luobenyan@zju.edu.cn

Abstract

Background and Objectives

Smoking cessation is universally prioritized for the prevention of cardiovascular disease and cancer, but its impact on dementia risk remains uncertain. We aimed to evaluate the associations of smoking cessation and postcessation weight gain with long-term risk of dementia and cognitive trajectories.

Methods

We conducted a prospective cohort study using data from the US Health and Retirement Study (1995–2020). A total of 32,802 dementia-free adults (mean age 60.5 years [SD 10.7]; 57.1% female) were included. Smoking status and body weight were assessed biennially through structured interviews. The primary outcome was incident dementia identified using the Langa-Weir algorithm, and the secondary outcome was cognitive function measured on a 27-point scale.

Results

Over 25 years of follow-up (median 9.9 years, interquartile range 4.4–16.4 years), 5,868 dementia cases were documented. Compared with current smokers, individuals who quit during follow-up had a lower dementia risk after quitting (hazard ratio 0.84, 95% CI 0.73–0.95), similar to those who had quit before baseline (0.79, 0.72–0.87) and to never smokers (0.75, 0.69–0.83). The benefits of cessation were largely limited to participants with no or modest 2-year postcessation weight gain (≤ 5 kg). By contrast, the association of quitting accompanied by >10 -kg weight gain was not statistically significant (1.33, 0.87–1.82). Restricted cubic spline analysis showed decreasing dementia risk with longer time since quitting, and the risk approached that of never smokers and plateaued at around 7 years after cessation. Cognitive trajectory analyses showed that quitting was associated with long-term slower cognitive decline (slope difference 0.19 points per decade, 95% CI 0.00–0.38) but no transient cognitive change (0.57; 95% CI –0.69 to 1.83), especially among those with minor weight gain (slope difference 0.23 per decade, 95% CI 0.03–0.43).

Discussion

Smoking cessation was associated with a sustained lower dementia risk and slower cognitive decline, comparable to never smokers and those without short-term risk increase. However, postcessation weight gain may attenuate these advantages, highlighting the need for weight management in cessation programs. These findings should be interpreted cautiously, given the potential residual confounding and measurement error.

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Supplementary Material

*These authors contributed equally to this work as co-corresponding authors.

¹Department of Neurology, First Affiliated Hospital of Zhejiang University School of Medicine, Hangzhou, China; ²School of Public Health, Second Affiliated Hospital, Zhejiang University School of Medicine, Hangzhou, China; ³Institute of Nutrition, Fudan University, Shanghai, China; ⁴Department of Nutrition, Harvard T. H. Chan School of Public Health, Boston, MA; and ⁵MOE Frontier Science Center for Brain Science & Brain-machine Integration, Zhejiang University, Hangzhou, China.

Glossary

AD = Alzheimer disease; BMI = body mass index; HR = hazard ratio; HRS = Health and Retirement Study.

Introduction

Dementia is a leading cause of disability and dependence among older adults, and its global prevalence is expected to rise sharply in the coming decades.^{1,2} Identifying modifiable risk factors is, therefore, a public health priority.^{3,4} Smoking is an established contributor to vascular damage and oxidative stress and increases the risk of cerebrovascular disease and diabetes, both of which are established risk factors of dementia.¹ While studies have suggested that current smokers are at a higher rate of cognitive decline and dementia, evidence on the cognitive effects of smoking cessation was rare and inconsistent.⁵⁻⁷ In these studies, former smokers experience risk reductions compared with current smokers, but the effect size varied.^{5,6}

A potential source of uncertainty emerged from the health consequences of weight gain after smoking cessation.^{8,9} Weight gain is a common postcessation outcome, potentially driven by increased appetite and a reduced basal metabolic rate associated with nicotine withdrawal.^{8,10} In a cohort study of approximately 170,000 participants, recent quitters had a 22% higher risk of type 2 diabetes compared with current smokers, with the risk peaking at 5–7 years after cessation and diminishing thereafter.¹¹ Crucially, the adverse metabolic effects were more pronounced in individuals who gained weight.^{11,12} Despite this short-term concern, the risk of mortality declined with cessation monotonically.^{11,13,14} These findings raise the question of whether postcessation weight gain may attenuate the cognitive benefits of quitting, as metabolic disorders are also associated with cognitive decline. Current studies on smoking cessation and dementia largely relied on single-point assessment of abstinence,^{5,15} which may underestimate the associations.¹⁶ In addition, while successful smoking quitters have more favorable brain structural changes,¹⁷ evidence on the role of postcessation weight gain in shaping dementia risk is scarce, especially during a long-term follow-up.

To address these gaps, we analyzed data from the Health and Retirement Study (HRS), a nationally representative cohort of US adults followed for over 2 decades. We hypothesized that smoking cessation would be associated with reduced dementia risk, but that excessive weight gain after cessation might attenuate or even offset these benefits.

Methods

Study Design and Participants

This study was based on HRS, a nationally representative longitudinal survey of US adults aged 50 years and older.¹⁸ The HRS, approved by the University of Michigan, collects detailed information on sociodemographic characteristics,

health behaviors, medical history, and cognitive function through biennial interviews. The design and procedures have been described in detail elsewhere.¹⁸

For this analysis, we included dementia-free participants from multiple subcohorts in the HRS since the 1995–1996 wave (eMethod 1), when cognitive assessments required for the Langa-Weir algorithm to define dementia first became available (eMethod 2). We excluded individuals with no follow-up interviews (eFigure 1), and the baseline characteristics of included vs excluded participants are given in eTable 1. Person-years were accumulated from the date of the baseline interview to dementia diagnosis, death, or the end of the study period (2020), whichever came first.

Assessment of Smoking Status and Weight Change

At each biennial interview, participants reported whether they had ever smoked and whether they were currently smoking. Based on these responses, we classified participants into 3 groups: current smokers, former smokers, and never smokers. During follow-up, individuals who reported smoking in the previous wave but not in the current wave were considered to have quit smoking.^{11,13} In the US population, self-reports of smoking are generally accurate and reliable.¹⁹ For quitters, we defined the onset of quitting as the beginning of the current wave and calculated the duration of smoking cessation until relapse, the occurrence of study outcomes, or the end of follow-up. Missing smoking information was imputed using responses from the previous wave.

Body weight was updated at every biennial interview. The validity of self-reported weight has been previously documented.^{20,21} We calculated 2-year weight changes among participants who had recently quit smoking (eFigure 2) to capture changes from precessation to postcessation without overlapping with subsequent follow-up. Postcessation weight gain was categorized as no weight gain, 0.1–5.0 kg, 5.1–10.0 kg, and >10.0 kg, consistent with previous literature.¹¹ These categories reflect clinically meaningful weight gain ranges and were selected to ensure adequate sample size and precision within each stratum. Because 2-year weight change was a key component of the exposure definition, participants with missing weight measurements during the relevant cessation interval were excluded from analyses involving postcessation weight change. To avoid potential exposure misclassification, weight change values were not imputed.

Assessment of Cognitive Function and Dementia

The primary outcome was incident dementia based on the Langa-Weir classification, which incorporated information

from objective measured cognitive function and proxy interviews.²² For self-respondents, the secondary outcome, cognitive function, was assessed using a validated 27-point scale, including immediate and delayed recall (0–20 points), serial sevens subtraction (0–5 points), and counting backward (0–2 points).²³ For participants unable to complete the cognitive battery, proxy interviews were used to generate a validated 11-point proxy scale combining informant memory ratings, instrumental activities of daily living, and interviewer assessments. Details are described in eMethods 2.

Covariates

In this study, analyses were adjusted for demographic, life-style, and health-related factors, consistent with previous research.^{6,11,24} Age was grouped in 5-year intervals, and study wave was included to account for period effects. Additional demographic covariates comprised sex (self-reported), marital status (married/partnered vs not), highest educational attainment (below high school, general educational development, high school graduate, some college, or college and above), and race (White, Black, or other). Lifestyle variables included physical activity (vigorous activity ≥ 3 times per week vs less), body mass index (BMI; <25.0 , 25.0 – 29.9 , or ≥ 30.0 kg/m²), and alcohol consumption (nondrinker, ≤ 1 drink per day, 1–2 drinks per day, or >2 drinks per day). Hypertension, diabetes, heart disease, and stroke were all self-reported. Given the low missing rates ($<10\%$) (eTable 2), we assigned a separate category to missing values.¹¹

Statistical Analysis

Baseline characteristics of participants were summarized according to smoking status at baseline. Continuous variables were presented as means with SDs and categorical variables as counts with percentages.

We used time-varying Cox proportional hazard regression to estimate hazard ratios (HRs) and 95% CIs of dementia by smoking status updated at each biennial wave. Smoking status was treated as a time-varying exposure and categorized using 3 hierarchical definitions: (1) current, former, and never smokers; (2) current smokers, quitters before baseline, quitters after baseline, and never smokers; and (3) current smokers, quitters before baseline, quitters without weight gain, quitters with a weight gain of 0.1–5.0, 5.1–10.0, or >10.0 kg, and never smokers. Smoking status was modeled as a time-varying variable. Participants were allowed to transition between categories over follow-up, including resumption of smoking after cessation. Accordingly, analyses based on different smoking definitions were conducted in separate models. In all analyses, current smokers served as the reference group. No violation of the proportional hazard assumption was detected for the associations using the Schoenfeld residual methods. The Cox models were stratified by age in 5-year groups and study wave, which allowed baseline hazards to vary across these strata to address confounding by age and period effects during the long-term follow-up. They were sequentially adjusted for sex, marital

status, highest educational attainment, race, physical activity, BMI category, alcohol drinking status, and the prevalence of hypertension, diabetes, heart disease, and stroke.

To assess the association between the duration of smoking cessation and dementia, we fitted restricted cubic spline models with knots placed at 5th, 50th, and 95th of cessation duration. We further tested for interaction between cessation duration and 2-year weight change categories to examine whether the trend differed by postcessation weight gain. We further performed prespecified subgroup analyses by age (≥ 70 years vs <70 years), sex, BMI category (<25 vs ≥ 25 kg/m²), educational attainment (college or above vs lower), and vigorous physical activity (≥ 3 times per week vs less). Interaction terms between each subgroup variable and the primary exposure were tested using likelihood ratio tests.

As a secondary analysis, we modeled cognitive trajectories using mixed-effects models with random intercepts and slopes to compare trajectories before and after smoking, and between quitters with ≥ 5.0 -kg weight gain and <5.0 -kg weight gain (eMethod 3).

We conducted several sensitivity analyses to examine the robustness of the findings by (1) excluding participants with diabetes, heart disease, stroke, or cancer at baseline; (2) excluding participants who developed dementia within the first 2 years since baseline; (3) excluding participants who were classified as cognitively impaired but not demented at baseline; (4) excluding participants who developed dementia within the first 2 years after smoking cessation; and (5) performing competing risk analysis using the Fine-Gray subdistribution hazard model, treating death as a competing event for dementia incidence.

All *p* values were 2-sided, with statistical significance defined as *p* < 0.05 . Analyses were conducted using R software, version 4.5.0 (R Foundation for Statistical Computing, Vienna, Austria).

Standard Protocol Approvals, Registrations, and Participant Consents

All HRS data collection protocols were approved by the University of Michigan Institutional Review Board. All HRS participants provided written informed consent at each wave of data collection. The HRS data used in this study are publicly available in fully deidentified form; therefore, no additional participant consent was required for this secondary data analysis.

Data Availability

Data are available in a public, open-access repository. The data used in this study were obtained from the HRS (hrs.isr.umich.edu), a publicly available longitudinal study sponsored by the National Institute on Aging (grant U01AG009740). The data set, data dictionary, and supporting documentation are accessible to registered users through the HRS website. Analytic codes used in this study are available from the corresponding authors on reasonable request.

Results

Characteristics of the Study Participants

Among the 32,802 participants, the mean age at baseline was 60.5 years (SD 10.7) and 57.1% were female. The cohort was predominantly White (73.4%), with Black participants comprising 18.1% and other races 8.6%. Among study participants, 30.5% had obesity. At study entry, 6,700 (20.4%) were current smokers, 11,951 (36.4%) were past smokers, and 14,151 (43.2%) had never smoked. Current smokers were younger (mean 57.2 years) than past smokers (62.2 years) and had lower educational attainment and household income. They are also less likely to have regular vigorous physical activity (32.9%). Alcohol drinking was more common and heavier among current and past smokers than among never smokers (Table 1). Detailed categories of weight change among participants who quit smoking during follow-up are presented in eTable 3.

Smoking Cessation, Weight Change, and Incident Dementia

Over a maximum of 25 years of follow-up (median 9.9 years, interquartile range 4.4–16.4 years), we documented 5,868 incident dementia cases (Table 2). In age-stratified and wave-stratified Cox models adjusted for sociodemographic factors, lifestyle, and comorbidities (model 3), smoking cessation was associated with a substantially lower risk of dementia compared with continued smoking. Compared with current smokers, both past smokers (HR 0.83, 95% CI 0.69–0.99) and never smokers (0.75, 0.69–0.83) had a lower risk of dementia. The associations were congruent when we further separate those who had quit before baseline and during the study follow-up: HR 0.84 (0.73–0.95) for participants who quit during follow-up and 0.79 (0.72–0.87) for those who had quit before baseline. The restricted cubic spline analysis showed decreasing dementia risk with longer time since quitting, and the risk approached those of never smokers and plateaued at around 7 years after cessation (Figure 1A). However, the 95% CIs after approximately 7 years of smoking cessation were relatively wide, indicating substantial uncertainty. This pattern should, therefore, be interpreted with caution.

When quitters were stratified by postcessation weight change, the cognitive benefits of quitting were confined to those with no or modest weight gain (p value <0.001 for interaction). Compared with current smokers, quitters with no weight gain had an adjusted HR of 0.83 (0.69–0.99), and quitters who gained 0.1–5.0 kg had an HR of 0.73 (0.60–0.90). By contrast, quitters who gained 5.1–10.0 kg (0.87, 0.66–1.15) did not have a significantly lower risk, and the HR for quitters with the largest gains (>10.0 kg) was 1.33 (0.97–1.82), which was not statistically significant. In the restricted cubic spline analysis, the risk of dementia was lower for quitters with no weight gain or a gain of 0.1–5.0 kg, but not among quitters who gained >5.0 kg (Figure 1B).

Smoking Cessation, Weight Change, and Cognitive Decline

Among the 3,419 participants who quit during follow-up and had repeated cognitive measurements, linear mixed-effects models showed no abrupt improvement in cognition immediately at the time of cessation (point estimate 0.57; 95% CI –0.69 to 1.83) but a slower subsequent rate of cognitive decline compared with participants who continued smoking (Table 3 and Figure 2A). Specifically, quitters experienced a change in the rate of decline of 0.19 points per decade (95% CI 0.00–0.38) compared with continuing smokers, indicating a modest attenuation of decline after quitting. In this cohort, each 10-year increase in age was associated with an average 1.20-point lower score on the 27-point cognitive scale. The observed slope difference, therefore, corresponds to an approximately 15% slower rate of age-related cognitive decline.

When stratified by postcessation weight change (Figure 2B), the slowing of decline was present and statistically significant only among quitters with no weight gain or a small gain of 0.1–5.0 kg (slope difference 0.23 per decade, 95% CI 0.03–0.43), which corresponds a 20% slower rate of cognitive decline. Quitters who gained >5.0 kg did not show a clear difference in slope compared with continuing smokers (change in slope 0.07, 95% CI –0.19 to 0.33).

Subgroup and Sensitivity Analyses

The association between smoke cessation and dementia was generally similar across study subgroups defined by age, BMI at baseline, and education attainment (eTable 4). However, the association was stronger among men (HR 0.70, 95% CI 0.57–0.86) compared with women (0.96, 0.81–1.15, p -interaction = 0.05), and stronger among quitters engaging in regular physical activity (0.76, 0.60–0.96) than those who did not (0.91, 0.77–1.07). Regarding weight gain after cessation, quitters who gained between 0.1 and 5.0 kg showed the most significant reduction in dementia risk, while those who gained more than 10 kg had a higher dementia risk across most subgroups, particularly in older participants and those with higher BMI (eTable 5).

The sensitivity analyses (eTable 6) generally supported the robustness of the associations between smoking cessation and dementia risk. Excluding participants with major chronic diseases at baseline, those who developed dementia within the first 2 years, or participants with baseline cognitive impairment, yielded results consistent with the primary analysis. When we excluded participants who developed dementia within the first 2 years after smoking cessation, the associations were strengthened: the HR for quitters who had no weight gain during follow-up was 0.46 (95% CI 0.36–0.58), stronger than that in the main analysis (0.83, 0.69–0.99). In addition, the competing risk analysis generated similar findings as in the primary analysis.

Table 1 Characteristics of Study Participants at Baseline

Characteristics	Overall population (N = 32,802)	By baseline smoking status		
		Current smokers (N = 6,700)	Past smokers (N = 11,951)	Never smokers (N = 14,151)
Age, y, mean (SD)	60.5 (10.7)	57.2 (8.4)	62.2 (10.7)	60.6 (11.3)
Female, n (%)	18,727 (57.1)	3,596 (53.7)	5,516 (46.2)	9,615 (67.9)
Married or partnered, n (%)	2,327 (70.9)	4,348 (64.9)	8,820 (73.8)	10,102 (71.4)
Race, n (%)				
White	24,067 (73.4)	4,543 (67.8)	9,253 (77.4)	10,271 (72.6)
Black	5,921 (18.1)	1,607 (24.0)	1,805 (15.1)	2,509 (17.7)
Other	2,814 (8.6)	550 (8.2)	893 (7.5)	1,371 (9.7)
Highest education attainment, n (%)				
Below high school	7,021 (21.4)	1,792 (26.7)	2,497 (20.9)	2,732 (19.3)
GED	1,634 (5.0)	542 (8.1)	648 (5.4)	444 (3.1)
High school graduate	9,393 (28.6)	1,985 (29.6)	3,329 (27.9)	4,079 (28.8)
Some college	7,869 (24.0)	1,684 (25.1)	2,944 (24.6)	3,241 (22.9)
College and above	6,883 (21.0)	696 (10.4)	2,533 (21.2)	3,654 (25.8)
Total annual household income, US dollars, n (%)				
<20,000	8,764 (26.7)	2,194 (32.7)	2,896 (24.2)	3,674 (26.0)
20,000–<40,000	7,640 (23.3)	1,672 (25.0)	2,943 (24.6)	3,025 (21.4)
40,000–<80,000	8,527 (26.0)	1,720 (25.7)	3,224 (27.0)	3,583 (25.3)
≥80,000	7,871 (24.0)	1,114 (16.6)	2,888 (24.2)	3,869 (27.3)
Body mass index, n (%)				
<25 kg/m ²	10,315 (31.4)	2,597 (38.8)	3,322 (27.8)	4,396 (31.1)
25–<30 kg/m ²	12,338 (37.6)	2,374 (35.4)	4,865 (40.7)	5,099 (36.0)
≥30 kg/m ²	9,563 (29.2)	1,657 (24.7)	3,584 (30.0)	4,322 (30.5)
Vigorous physical activity ≥3 times per week, n (%)	12,938 (39.4)	2,206 (32.9)	5,010 (41.9)	5,722 (40.4)
Alcohol intake, n (%)				
Nondrinker	20,152 (61.4)	3,592 (53.6)	6,744 (56.4)	9,816 (69.4)
≤1 drink per day	8,749 (26.7)	1,731 (25.8)	3,589 (30.0)	3,429 (24.2)
1–2 drinks per day	2,319 (7.1)	695 (10.4)	1,028 (8.6)	596 (4.2)
>2 drinks per day	1,444 (4.4)	629 (9.4)	546 (4.6)	269 (1.9)
Self-reported health conditions, n (%)				
High blood pressure	13,030 (39.7)	2,499 (37.3)	4,973 (41.6)	5,558 (39.3)
Diabetes	4,303 (13.1)	801 (12.0)	1,774 (14.8)	1,728 (12.2)
Heart diseases	4,540 (13.8)	814 (12.1)	2,057 (17.2)	1,669 (11.8)
Stroke	1,379 (4.2)	326 (4.9)	587 (4.9)	466 (3.3)

Abbreviation: GED = general educational development.

Baseline characteristics are presented according to smoking status at the first study observation (baseline). Although smoking status was treated as a time-varying exposure in the main analyses, categories in this table reflect baseline status only. “Current smoker” indicates active smoking at baseline and does not imply continued smoking during follow-up. Percentages for categorical variables may not sum to 100% because of missing data, which were handled using a separate indicator category in the formal analysis.

Table 2 HRs of Dementia by Smoking Cessation Status

Smoking status	Cases/person-years	HR (95% CI)		
		Model 1	Model 2	Model 3
Current smokers	702/46,524	1 (ref.)	1 (ref.)	1 (ref.)
Past smokers	2,556/160,192	0.63 (0.58–0.69)	0.80 (0.73–0.87)	0.80 (0.73–0.88)
Quitters, before baseline	2,198/137,626	0.61 (0.55–0.66)	0.81 (0.74–0.89)	0.79 (0.72–0.87)
Quitters, during follow-up	358/22,566	0.81 (0.71–0.93)	0.85 (0.74–0.97)	0.84 (0.73–0.95)
No weight gain	154/9,054	0.86 (0.72–1.03)	0.84 (0.71–1.01)	0.83 (0.69–0.99)
Weight gain of 0.1–5.0 kg	107/7,846	0.68 (0.55–0.83)	0.74 (0.60–0.91)	0.73 (0.60–0.90)
Weight gain of 5.1–10.0 kg	55/3,551	0.78 (0.59–1.03)	0.86 (0.65–1.14)	0.87 (0.66–1.15)
Weight gain of >10.0 kg	42/2,019	1.29 (0.94–1.77)	1.40 (1.02–1.93)	1.33 (0.97–1.82)
Never smokers	2,610/162,767	0.60 (0.55–0.66)	0.76 (0.69–0.83)	0.75 (0.69–0.83)

Abbreviations: GED = general educational development; HR = hazard ratio. Smoking status was treated as a time-varying exposure and categorized using 3 hierarchical definitions: (1) current, former, and never smokers; (2) current smokers, quitters before baseline, quitters after baseline, and never smokers; and (3) current smokers, quitters before baseline, quitters without weight gain, quitters with a weight gain of 0.1–5.0, 5.1–10.0, or >10.0 kg, and never smokers. Each smoking classification corresponds to a separately fitted Cox proportional hazard model. In all analyses, current smokers served as the reference group. Participants could contribute person-time to more than 1 category across follow-up. Therefore, this table reports numbers of cases and accumulated person-years for each category, corresponding to the risk sets used in the Cox models, rather than mutually exclusive participant counts. HRs and 95% CIs were estimated using Cox proportional hazard models. Model 1 was the age-stratified and wave-stratified Cox models to account for the confounding of age (categorized into 5-year groups) and study wave (period effect). Model 2 was further adjusted for sex (male or female), marital status (married/partnered or not), education (below high school, GED, high school graduate, some college, college and above), race (White, Black or other), total household income (<20,000, 20,000–40,000, 40,000–80,000, >80,000 US dollars), physical activity (vigorous physical activity no less than 3 times per week or not), baseline body mass index (<25, 25–30, >30 kg/m²), and alcohol intake (never, 0–1, 1–2, >2 drinks per day). Model 3 was based on model 2 and further adjusted for comorbidities: hypertension, diabetes, heart disease, and stroke.

Discussion

In this prospective study, smoking cessation was associated with a lower risk of dementia, with no sudden impact on cognitive function or elevated dementia risk. The cognitive benefits were confined to individuals with minimal-to-modest postcessation weight gain, while substantial weight gain may attenuate these advantages. These findings highlight the importance of incorporating weight management strategies in smoking cessation programs.

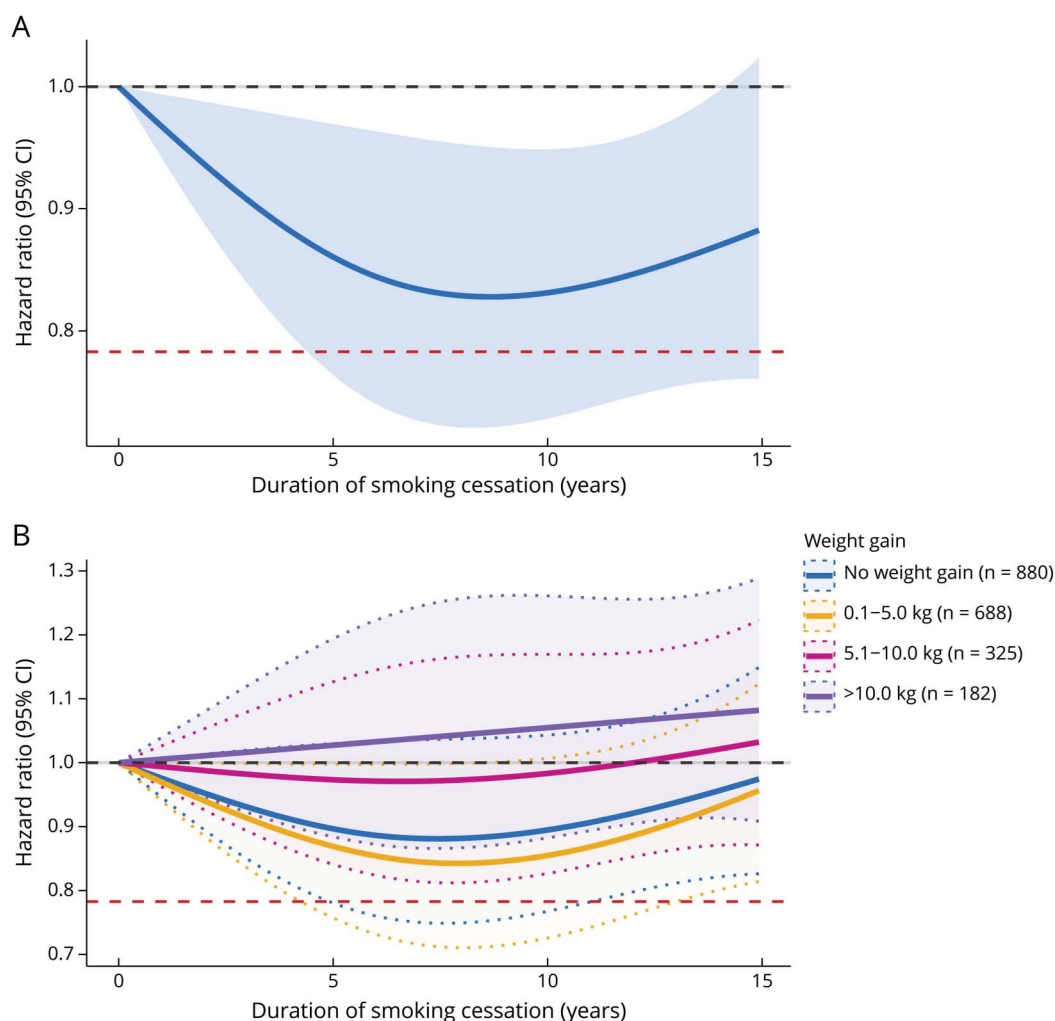
The observed association of smoking cessation with a lower dementia risk accords with previous studies suggesting that smoking is a major risk factor of cognitive decline and dementia. According to a meta-analysis, current smokers have a 30% higher risk of all-cause dementia, 40% higher risk of Alzheimer disease (AD), and 38% higher risk of vascular dementia.²⁵ In a recent meta-analysis, smoking accounted for 4.5% of the incidence of dementia worldwide.²⁶ In our study, both former smokers and those who quit during the follow-up period had a significantly lower risk of dementia compared with current smokers, with the greatest benefit observed in those who quit smoking long term. These findings align with the growing body of literature supporting the cognitive benefits of smoking cessation, with sustained cessation over time. In a cohort study in Japan (mean follow-up 5.7 years), current smokers had a higher risk of incident dementia compared with never smokers, but ex-smokers who maintained abstinence for at least 3 years (retrospectively collected status) showed

similar risk compared with never smokers.¹⁵ Similarly, in a 7-year Korean cohort study, long-term quitters and never smokers at baseline had a lower risk of dementia.⁵ It is important to note that we did not observe a short-term elevated dementia risk in former smokers, which echoed previous evidence on no short-term elevated risks of mortality, despite increased diabetes risk in the first 5–7 years.¹¹

The role of postcessation weight gain in brain health remains largely unclear. Based on the previous studies on the metabolic risks associated with postcessation weight gain,¹¹ we found that a postcessation weight gain of 0.1–5.0 kg was associated with further reductions in dementia risk compared with continuing smoking. However, individuals who gained >5.0 kg experienced no significant cognitive benefit. This suggests that the overall benefits of smoking cessation do not necessarily outweigh the harms of substantial weight gain, and that excessive weight gain after cessation may compromise the long-term cognitive advantages. Public health interventions should, therefore, not only encourage smoking cessation but also integrate weight management to prevent excessive weight gain.

The relationship between weight gain and cognitive outcomes after smoking cessation is further supported by our analysis of cognitive decline trajectories. Quitters who gained little to no weight experienced approximately 20% slower rates of cognitive decline compared with continuing smokers without sudden impact, which is meaningful at the population level, given the high prevalence of smoking and dementia. By

Figure 1 Restricted Cubic Splines for Association Between Duration of Smoking Cessation and Risk of Dementia



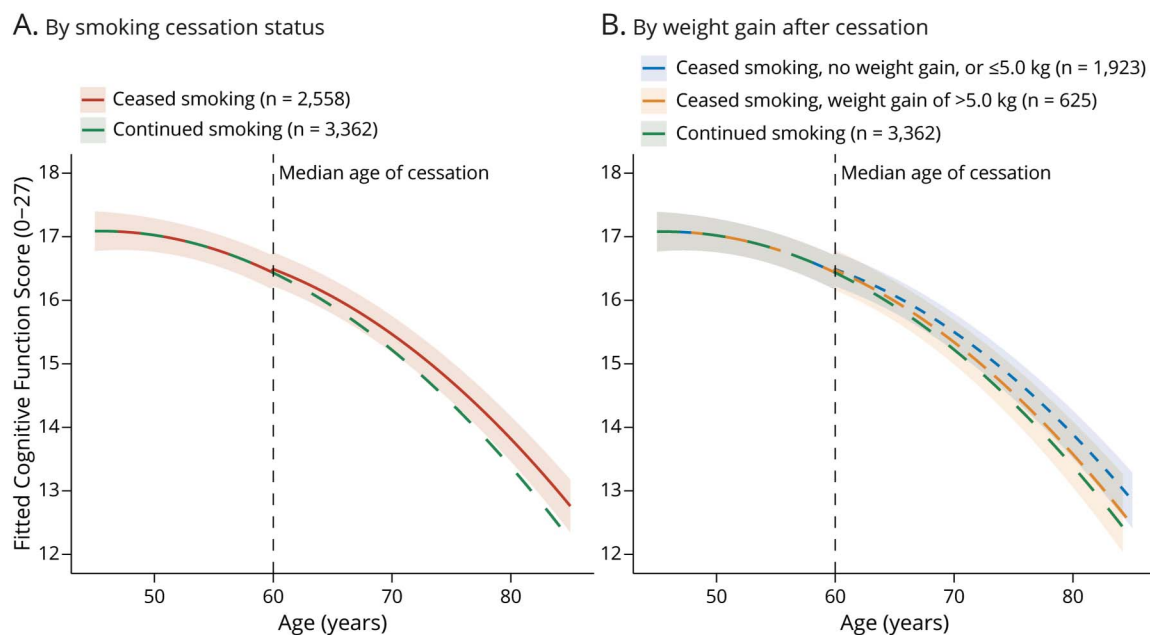
Panel A shows the estimated hazard ratios and 95% CIs (shaded area) of dementia risk according to the duration of smoking cessation, and panel B shows the hazard ratios by postcessation weight gain, both modeled using restricted cubic splines with 4 knots. The red dashed line indicated the hazard ratio for participants who never smoked, while the black dashed line indicated hazard ratio for current smokers (the reference group). The restricted cubic splines were fitted within a Cox proportional hazard model stratified by age (5-year groups) and study wave and adjusted for sex, marital status, highest education attainment, race, physical activity, BMI category, alcohol drinking status, and the prevalence of hypertension, diabetes, heart diseases, and stroke.

Table 3 Difference in Trajectories of Cognitive Change per Decade Before and After Cessation Among Participants Quitting Smoking During Follow-Up

Variable	Sudden change in cognitive function	Change in rate of cognitive decline with linear age
By smoking status		
Current smokers	0 (ref.)	0 (ref.)
Quitters	0.57 (−0.69 to 1.83)	0.19 (0.00 to 0.38)
By weight gain after cessation		
Current smokers	0 (ref.)	0 (ref.)
Quitters with no or 0.1–5.0 kg weight gain	0.54 (−0.82 to 1.91)	0.23 (0.03 to 0.43)
Quitters with weight gain of >5.0 kg	0.44 (−1.67 to 2.56)	0.07 (−0.19 to 0.33)

Estimates are from linear mixed-effects models with random intercepts, adjusted for demographic, lifestyle, and health-related covariates. Sudden change reflects the immediate shift in cognition at cessation and changes in rate of decline reflects the difference in slope of cognitive decline per decade compared with current smokers. Details of the models are described in eMethod 3.

Figure 2 Fitted Trajectories of Cognitive Function by Smoking Cessation Status



Panel A shows the fitted cognitive trajectories and 95% CIs (shaded area) of a hypothetical person with smoking cessation at 60 years of age (population median age at cessation, presented as the vertical lines) in the reference categories of covariates (White race, female, married or partnered, 12 years of education, vigorous physical activity >3 times per week, underweight or normal weight, no history of alcohol use, and no baseline comorbidities) compared with cognitive trajectories of a person with the same characteristics who remained smoking. Panel B shows the fitted the cognitive trajectories by weight gain around smoking cessation.

contrast, those who gained significant weight (>5.0 kg) did not show an improvement in cognitive decline, reinforcing the notion that excessive weight gain may negate some of the cognitive benefits associated with smoking cessation.

The mechanisms by which smoking accelerates neurodegeneration have been widely studied, with evidence linking smoking to vascular damage, oxidative stress, and inflammation, all of which may contribute to cognitive decline and dementia.²⁷ In addition, smoking may exacerbate key pathologic features of AD, such as amyloid plaque formation, neuroinflammation, and tau phosphorylation.^{28,29} However, the potential mechanisms of the adverse effect of excessive weight gain after smoking cessation warrants future investigations.³⁰

Strengths of this study include its large, nationally representative sample and an extended follow-up period of up to 25 years, providing a robust basis for examining the long-term effects of smoking cessation on dementia risk. The repeated assessment of smoking status, weight change, and dementia outcomes, along with comprehensive adjustments for a wide range of potential confounding factors, allowed for a detailed analysis of smoking cessation trajectories during follow-up, the duration of smoking cessation, and postcessation weight changes. However, several limitations should be noted. First, the use of self-reported smoking and weight data may introduce recall bias, although the large sample size and long follow-up period mitigate some of these concerns. Second, the study design is observational, so causality cannot be

definitively established. Although we have controlled for a range of potential confounders, unmeasured factors may still influence the observed associations. In addition, the classification of smoking cessation based on self-reported smoking status may not capture the full complexity of smoking patterns, such as short-term intermittent smoking or relapse within the 2-year intervals, which could affect the outcomes. Weight gain after smoking cessation may, in part, reflect early metabolic or behavioral changes associated with preclinical neurodegeneration. Alterations in appetite regulation, energy expenditure, metabolic health, and physical activity have been observed in the prodromal phase of dementia and could contribute to both postcessation weight change and subsequent dementia risk. Moreover, progressive weight loss is more commonly observed in the preclinical phase of dementia, which may have attenuated the observed associations for smoking cessation with weight loss. Finally, although this study involves a wide range of races and ethnicities, whether the study findings may be generalized to other populations warrants investigation. Differences in health care infrastructure, nutritional environments, and social determinants of health across settings may modify these associations. Future studies in more socioeconomically and ethnically diverse populations are needed to evaluate the applicability of these findings and their implications for health disparities.

Our findings emphasize the importance of smoking cessation for dementia prevention among smokers, and that excessive weight gain after cessation may attenuate or even negate the

cognitive benefits. These results suggest that public health interventions aimed at reducing smoking-related dementia risk should consider not only the promotion of smoking cessation but also weight management with quitting to maximize the health benefits. Future research should further explore the mechanisms by which weight gain may influence brain health and investigate potential interventions to support both smoking cessation and weight management in aging populations.

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Author Contributions

H. Chen: drafting/revision of the manuscript for content, including medical writing for content; study concept or design; analysis or interpretation of data. J. Wang: drafting/revision of the manuscript for content, including medical writing for content; analysis or interpretation of data. S. Lai: drafting/revision of the manuscript for content, including medical writing for content; analysis or interpretation of data. G. Peng: drafting/revision of the manuscript for content, including medical writing for content. G. Zong: drafting/revision of the manuscript for content, including medical writing for content; study concept or design. C. Yuan: drafting/revision of the manuscript for content, including medical writing for content; study concept or design. B. Luo: drafting/revision of the manuscript for content, including medical writing for content; study concept or design; analysis or interpretation of data.

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Disclosure

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